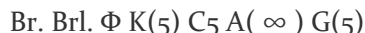


5. A Rose by another Name (20 marks)

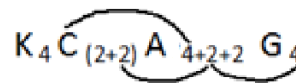
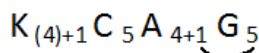
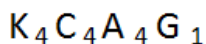
Problem designed by Abhishek Dedhe

You must have heard about chemical formulas, but did you know that there are also *floral formulas* for representing the structure of flowers? For example, the floral formula of *China Rose* is:



Don't worry if that looks bizarre. Not all flowers have such complicated formulas. In fact, there is a systematic way in which the structure of a flower (for example, whether it is symmetrical, how many petals does it have, how are the petals connected to each other, and so on) can be represented in terms of a *floral diagram*. Since drawing a detailed diagram can take time, botanists have invented a clever way of representing the diagrams in terms of floral formulas.

Given below are some floral diagrams and their corresponding floral formulas.



Assignment 1: Explain the rules of obtaining the *floral formula* from a *floral diagram*.

Assignment 2 [Answer-booklet]: Given below are three floral diagrams. Write their corresponding floral formulae.



(a)



(b)



(c)

Assignment 3: Explain the problem(s) which would arise when trying to draw the floral diagram of this floral formula: $\text{K}_{(6)}\text{C}_{(2+2)}\text{A}_4\text{G}_2$ How might you resolve this issue?

Note: The data presented in this problem has been slightly simplified.